

Software Requirements Specification (SRS)

Internet Banking System

Document Version: 1.0

Standard Followed: IEEE 830-1998 / IEEE 29148-2018

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Organization: University

Project Type: Real Banking Solution (Bangladesh Context)

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1. Introduction

1.1 Purpose

এই SRS Document এর উদ্দেশ্য হলো **Internet Banking System** এর সম্পূর্ণ software requirement সংজ্ঞায়িত করা, যা development team, stakeholders এবং auditors এর মধ্যে একটি authoritative reference হিসেবে কাজ করবে। এটি IEEE 29148 standard অনুসরণ করে প্রস্তুত।

1.2 Scope

Product Name: Internet Banking System (IBS)

Version: 1.0

In-Scope:

- Customer online banking portal (web-based)
- Admin/Employee back-office panel
- Core banking operations (account, transfer, bill, card, loan)
- AI-based fraud detection, chatbot, KYC OCR
- Azure cloud deployment

Out-of-Scope (Phase 1):

- Mobile native app (Android/iOS) — future phase
- ATM/POS integration
- Investment/stock trading
- International remittance

Benefits:

- 24/7 self-service banking
- Reduced branch load
- Real-time fraud detection
- Regulatory compliance (Bangladesh Bank)

Objectives:

- Reduce transaction cost by 40%
- Achieve 99.9% uptime
- Detect fraud within 3 seconds
- Support 100,000+ concurrent users

1.3 Definitions, Acronyms, Abbreviations

Term	Meaning
IBS	Internet Banking System
KYC	Know Your Customer
NID	National ID (Bangladesh)
2FA	Two-Factor Authentication
JWT	JSON Web Token
OTP	One-Time Password
AML	Anti-Money Laundering
CFT	Combating Financing of Terrorism
BB	Bangladesh Bank
PCI-DSS	Payment Card Industry Data Security Standard
SRS	Software Requirements Specification
REST	Representational State Transfer
SLA	Service Level Agreement
RTO	Recovery Time Objective
RPO	Recovery Point Objective

1.4 References

1. IEEE Std 830-1998 – Recommended Practice for SRS
2. IEEE Std 29148-2018 – Requirements Engineering
3. Bangladesh Bank ICT Security Guideline (2023)
4. Bangladesh Bank Guidelines on ICT Risk Management
5. PCI-DSS v4.0
6. ISO/IEC 27001:2022
7. GDPR (for future international expansion)

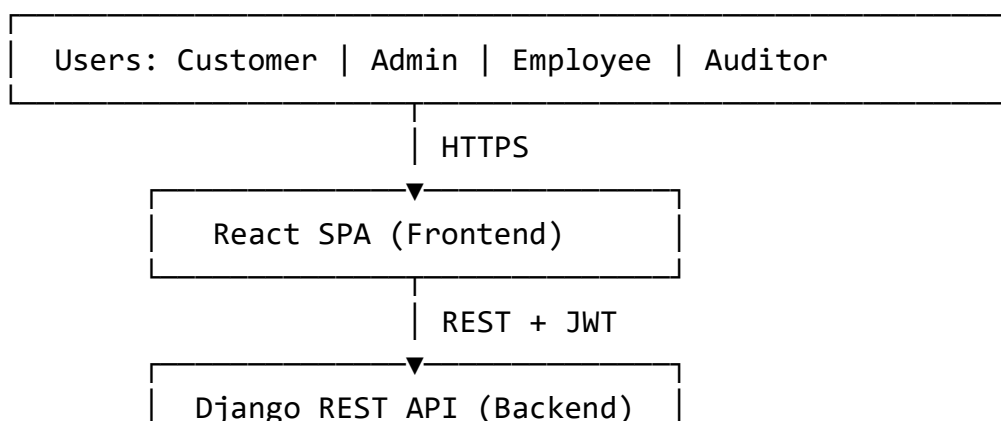
1.5 Overview

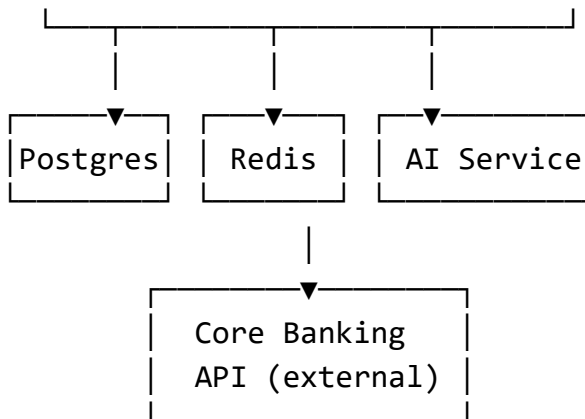
Section 2 এ overall description, Section 3-4 এ functional/non-functional requirement, Section 5-7 এ architecture/data/interface, Section 8-11 এ AI, security, testing, deployment বর্ণনা করা হয়েছে।

2. Overall Description

2.1 Product Perspective

IBS হলো একটি **three-tier web application** যা bank এর core banking system এর সাথে API integration করে। এটি একটি **greenfield real project**, যেখানে modern microservice architecture, cloud-native deployment, এবং AI-powered features ব্যবহার করা হবে।





2.2 Product Functions (High-Level)

#	Function	Actor
F1	Registration & KYC	Customer
F2	Login with 2FA/OTP	All users
F3	Balance & Statement	Customer
F4	Fund Transfer	Customer
F5	Beneficiary Management	Customer
F6	Bill Payment	Customer
F7	Card Management	Customer
F8	Loan Application	Customer
F9	Complaint/Ticket	Customer
F10	KYC Approval	Admin
F11	Transaction Monitoring	Admin/Employee
F12	Report Generation	Admin/Auditor

F13	AI Fraud Alert	System → Admin
F14	AI Chatbot Support	Customer
F15	Audit Trail	System

2.3 User Classes & Characteristics

User Class	Frequency	Skill Level	Privileges
Customer	Daily	Basic	Own accounts only
Bank Employee	Daily	Intermediate	Read customer info, resolve ticket
Bank Admin	Daily	Advanced	Full customer mgmt, KYC, limits
Auditor	Weekly	Advanced	Read-only audit logs & reports
System Admin	On-demand	Expert	Infra, deployment, backup
AI Service	Continuous	N/A	Programmatic access

2.4 Operating Environment

Client:

- Modern browser (Chrome 110+, Firefox 110+, Edge 110+, Safari 16+)
- Min screen 360×640, optimum 1366×768+
- Internet: min 512 kbps

Server:

- Linux (Ubuntu 22.04 LTS)
- Python 3.11+, Node 20+
- PostgreSQL 16, Redis 7
- Azure App Service / Container Apps

2.5 Design & Implementation Constraints

- **CO-01:** Bangladesh Bank ICT Security Guideline মেনে চলতে হবে
- **CO-02:** সব data Bangladesh এ hosted থাকতে হবে (data sovereignty)
- **CO-03:** PCI-DSS compliance card data এর জন্য
- **CO-04:** TLS 1.3 mandatory
- **CO-05:** Password minimum 12 characters
- **CO-06:** AI model explainability থাকতে হবে (audit)
- **CO-07:** Backup daily, retention 7 years (BB guideline)
- **CO-08:** Support Bangla & English UI

2.6 Assumptions & Dependencies

Assumptions:

- A-01: Customer এর কাছে valid mobile number ও NID আছে
- A-02: Bank এর core banking system API available
- A-03: Internet connectivity (customer side)
- A-04: SMS/Email gateway (SSL Wireless/Twilio) available

Dependencies:

- D-01: Core Banking System REST API
- D-02: NID verification API (Election Commission / Porichoy)
- D-03: Payment gateway (bKash/Nagad/SSLCommerz)
- D-04: Azure cloud subscription
- D-05: SSL certificate

2.7 Apportioning of Requirements

Phase	Features
Phase 1 (MVP)	Registration, Login, Balance, Transfer, Statement, Admin panel
Phase 2	Bills, Cards, Loans, Notifications

Phase 3	AI Fraud, Chatbot, OCR KYC
Phase 4	Mobile app, Investment, Remittance

3. Functional Requirements

3.1 User Management (FR-UM)

FR-UM-01: User Registration

- **Description:** Customer নিজে register করতে পারবে email, phone, NID দিয়ে।
- **Input:** Full name, email, phone, password, NID number
- **Process:** Validate → Duplicate check → Hash password → Save (status=PENDING) → Send OTP
- **Output:** Registration success message + OTP
- **Priority:** Must
- **Acceptance:** 100-character name, valid email, 11-digit BD phone, 10/17-digit NID

FR-UM-02: Email Verification

FR-UM-03: Login with Password

FR-UM-04: Two-Factor Authentication (OTP)

FR-UM-05: Password Reset

FR-UM-06: Profile Update

FR-UM-07: Account Block/Unblock (Admin)

3.2 KYC Management (FR-KYC)

FR-KYC-01: KYC Submission

- NID copy (front/back), photo, address proof upload
- Auto OCR extraction

FR-KYC-02: KYC Approval Workflow

PENDING → [Admin review] → APPROVED / REJECTED
|
└ REJECTED: customer re-submit

FR-KYC-03: KYC Re-verification (yearly)

3.3 Account Management (FR-ACC)

FR-ACC-01: View Account Balance

FR-ACC-02: Account Statement (Mini + Full PDF)

FR-ACC-03: Multiple Accounts Support

FR-ACC-04: Account Type Info

3.4 Beneficiary Management (FR-BEN)

FR-BEN-01: Add Beneficiary

FR-BEN-02: Beneficiary Verification (IBAN/Account check)

FR-BEN-03: Edit/Delete Beneficiary

FR-BEN-04: Beneficiary Limit Setup

3.5 Fund Transfer (FR-TXN)

FR-TXN-01: Own Account Transfer

FR-TXN-02: Other Account (Within Bank)

FR-TXN-03: Other Bank Transfer (BEFTN/RTGS)

FR-TXN-04: Transfer with OTP Verification

FR-TXN-05: Scheduled/Recurring Transfer

FR-TXN-06: Transaction Reversal (Admin)

Business Rules:

- BR-01: Daily limit customer: ₺5,00,000
- BR-02: Per transaction limit: ₺2,00,000
- BR-03: OTP expire: 5 minutes
- BR-04: 3 failed OTP → account lock 30 min
- BR-05: Fraud score > 0.8 → hold for manual review

3.6 Bill Payment (FR-BILL)

FR-BILL-01: Biller List

FR-BILL-02: Bill Payment

FR-BILL-03: Bill History

FR-BILL-04: Auto-bill reminder

3.7 Card Management (FR-CARD)

FR-CARD-01: View Cards

FR-CARD-02: Block/Unblock Card

FR-CARD-03: Set Daily Limit

FR-CARD-04: Card Statement

3.8 Loan Management (FR-LOAN)

FR-LOAN-01: Loan Application

FR-LOAN-02: Eligibility Check (AI-powered)

FR-LOAN-03: EMI Calculator

FR-LOAN-04: Approval Workflow

FR-LOAN-05: Loan Payment

3.9 Notification (FR-NOTIF)

FR-NOTIF-01: Email Notification

FR-NOTIF-02: SMS Notification

FR-NOTIF-03: In-App Notification

FR-NOTIF-04: Notification Preferences

3.10 Complaint/Ticket (FR-CMP)

FR-CMP-01: Submit Complaint

FR-CMP-02: Ticket Tracking

FR-CMP-03: Ticket Assignment (Admin)

FR-CMP-04: Ticket Resolution

3.11 Admin Functions (FR-ADM)

FR-ADM-01: User Management

FR-ADM-02: Transaction Monitoring

FR-ADM-03: Report Generation

FR-ADM-04: Limit Configuration

FR-ADM-05: System Settings

3.12 Audit (FR-AUD)

FR-AUD-01: Log Every Action

FR-AUD-02: Immutable Audit Log

4. Non-Functional Requirements

4.1 Performance (NFR-PERF)

ID	Requirement	Target
NFR-PERF-01	Page load time	< 2s (P95)
NFR-PERF-02	API response	< 500ms (P95)
NFR-PERF-03	Fund transfer completion	< 3s
NFR-PERF-04	Concurrent users	100,000+
NFR-PERF-05	TPS (transactions/sec)	5,000 peak
NFR-PERF-06	AI fraud check	< 1s

4.2 Scalability (NFR-SCAL)

- Horizontal scaling via containers
- DB read replicas
- Redis caching
- Auto-scaling (Azure)

4.3 Availability (NFR-AVAIL)

ID	Requirement
NFR-AVAIL-01	Uptime 99.9% (excluding planned maintenance)
NFR-AVAIL-02	Planned downtime < 4 hours/month (2 AM–4 AM)

NFR-AVAIL-03	RTO: 1 hour
NFR-AVAIL-04	RPO: 15 minutes

4.4 Security (NFR-SEC)

ID	Requirement
NFR-SEC-01	TLS 1.3 for all traffic
NFR-SEC-02	Password: PBKDF2/bcrypt/Argon2 (min 12 chars)
NFR-SEC-03	JWT access token: 30 min, refresh: 7 days
NFR-SEC-04	Rate limiting: 5 login attempts/min
NFR-SEC-05	AES-256 for PII at rest
NFR-SEC-06	SQL injection prevention (ORM)
NFR-SEC-07	XSS/CSRF protection
NFR-SEC-08	Secrets in Azure Key Vault
NFR-SEC-09	Session invalidation on logout
NFR-SEC-10	IP whitelisting for admin panel

4.5 Usability (NFR-USE)

- Bangla & English bilingual UI
- Mobile responsive (Bootstrap 5)
- WCAG 2.1 AA compliance
- Max 3 clicks for common tasks
- Contextual help

4.6 Reliability (NFR-REL)

- Transaction atomicity (ACID)
- Idempotency keys for transfers
- Auto-retry with backoff
- Circuit breaker for external APIs

4.7 Maintainability (NFR-MAIN)

- Modular code (Django apps)
- 80% test coverage
- API docs via Swagger/OpenAPI
- CI/CD automated

4.8 Compliance (NFR-COMP)

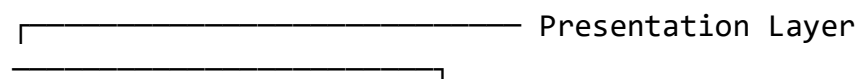
- Bangladesh Bank ICT Security Guideline
- PCI-DSS v4.0 (card data)
- ISO 27001
- AML/CFT rules
- Data retention: 7 years

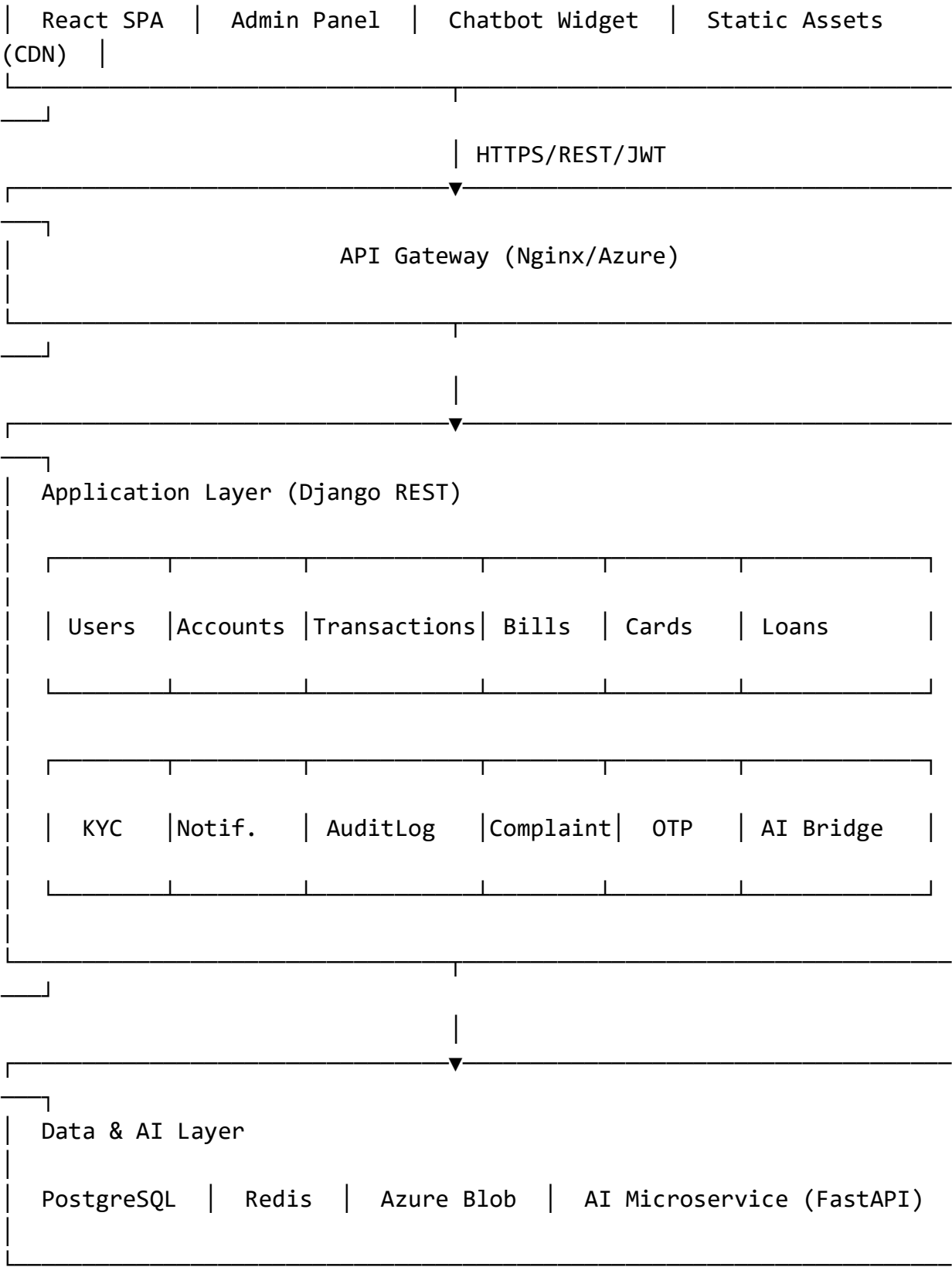
4.9 Portability (NFR-PORT)

- Docker container
- Cloud-agnostic (Azure primary, AWS fallback)
- Multi-OS dev support

5. System Architecture

5.1 Logical Architecture





5.2 Technology Stack

Layer	Technology	Version
Frontend	React	18
UI	Bootstrap + React-Bootstrap	5.3
Backend	Django + DRF	5.0 / 3.15
Auth	djangorestframework-simplejwt	5.3
DB	PostgreSQL	16
DB Admin	pgAdmin4	8
Cache	Redis	7
AI	Python + FastAPI + scikit-learn	3.11 / 0.109
Container	Docker	25
Cloud	Microsoft Azure	—
CI/CD	GitHub Actions	—
Monitoring	Azure App Insights	—

5.3 Deployment Architecture (Azure)

Component	Azure Service
Frontend	Azure Static Web Apps
Backend	Azure App Service (Linux)

AI Service	Azure Container Apps
Database	Azure DB for PostgreSQL Flexible Server
Cache	Azure Cache for Redis
Storage	Azure Blob Storage
Secrets	Azure Key Vault
CDN	Azure CDN
DNS	Azure DNS
Monitoring	Application Insights
Backup	Azure Backup

6. Data Model

6.1 Entity Relationship (Summary)

Entity	Key Attributes	Relations
users	user_id PK, email, phone, role, status	1:N → accounts, beneficiaries, ...
kyc	kyc_id PK, user_id FK, nid, status	1:1 → users
accounts	account_id PK, user_id FK, balance	1:N → transactions, cards
beneficiaries	beneficiary_id PK, user_id FK	1:N → transactions
transactions	txn_id PK, from_acc, to_acc, amount, status	N:1 → accounts
bills	bill_id PK, user_id FK, biller, amount	1:1 → transactions

cards	card_id PK, account_id FK, type	N:1 → accounts
loans	loan_id PK, user_id FK, principal	N:1 → users
notifications	notif_id PK, user_id FK, channel	N:1 → users
audit_logs	log_id PK, user_id FK, action	N:1 → users
complaints	ticket_id PK, user_id FK, status	N:1 → users
otp_verifications	otp_id PK, user_id FK, purpose	N:1 → users

(Full schema Section 2.4 আগের উত্তরে দেওয়া হয়েছে)

6.2 Data Volume Estimate

Table	Rows/Year	Growth
users	500,000	+20%/yr
accounts	700,000	+25%/yr
transactions	50,000,000	+30%/yr
audit_logs	100,000,000	+40%/yr
notifications	20,000,000	+35%/yr

6.3 Backup & Retention

Type	Frequency	Retention
Full DB backup	Daily 2 AM	30 days
Incremental	Every 15 min	7 days
Archive	Monthly	7 years (BB rule)

Blob (KYC docs)	Real-time	10 years
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7. External Interface Requirements

7.1 User Interfaces

- **UI-01:** Login page — email/phone + password + OTP
- **UI-02:** Dashboard — balance cards, chart, quick actions
- **UI-03:** Transfer wizard — 4-step (beneficiary → amount → review → OTP)
- **UI-04:** Admin dashboard — users, transactions, alerts
- **UI-05:** Chatbot floating widget (bottom-right)
- **UI-06:** Bangla/English toggle

UI Standards:

- Bootstrap 5 components
- Color: Navy #003366, Accent: Gold #FFB800
- Font: Inter / Noto Sans Bengali
- Logo top-left, user menu top-right

7.2 Hardware Interfaces

- Server: 4 vCPU, 16 GB RAM (min)
- Load balancer: Azure LB
- Storage: SSD, 500 GB+

7.3 Software Interfaces

External System	Purpose	Protocol
Core Banking System	Account/Transaction sync	REST/SOAP
NID Verification (Porichoy)	KYC validation	REST

SMS Gateway (SSL Wireless)	OTP, alerts	REST
Email (SendGrid/Azure)	Notifications	SMTP/API
Payment Gateway (SSLCommerz)	Bill payment	REST
Azure Blob	File storage	SDK
AI Microservice	Fraud/Chatbot/OCR	REST

7.4 Communication Interfaces

- HTTPS (TLS 1.3)
- JSON payload
- REST endpoints: `/api/v1/...`
- WebSocket (future: real-time notifications)
- Time format: ISO 8601 UTC

8. AI Module Requirements

8.1 Fraud Detection (AI-FRD)

AI-FRD-01: প্রতিটি transaction এ real-time fraud score (0–1)

AI-FRD-02: Score > 0.8 → transaction hold + admin alert

AI-FRD-03: Model: Isolation Forest / XGBoost

AI-FRD-04: Features: amount, hour, txn_type, device_change, location_change, velocity (txn/hour), merchant_category

AI-FRD-05: Retraining: monthly (with labeled data)

AI-FRD-06: Explainability: SHAP values in admin panel

AI-FRD-07: Accuracy ≥ 95%, FPR ≤ 2%

8.2 Chatbot (AI-CHAT)

AI-CHAT-01: 24/7 support (Bangla + English)

AI-CHAT-02: Handles: balance query, mini statement, branch info, complaint status

AI-CHAT-03: Fallback to human agent if confidence < 0.6

AI-CHAT-04: No sensitive action (transfer) without OTP re-auth

8.3 KYC OCR (AI-OCR)

AI-OCR-01: Auto-extract NID number, name, DOB from uploaded image

AI-OCR-02: Confidence threshold 0.85 → else manual review

AI-OCR-03: Detect fake/tampered documents

AI-OCR-04: Engine: EasyOCR / Tesseract + CNN classifier

8.4 Credit Scoring (AI-CREDIT)

AI-CREDIT-01: Loan eligibility score (0–100)

AI-CREDIT-02: Features: income, existing loans, repayment history, account age

AI-CREDIT-03: Model: Logistic Regression / Random Forest

AI-CREDIT-04: Provide reason codes for rejection

8.5 AI Ethics & Governance

- Model bias audit quarterly
- No discrimination based on gender/religion/location
- Human-in-the-loop for rejections
- Model versioning & rollback support
- Data privacy: PII hashed before training

9. Security & Compliance

9.1 Authentication & Authorization

- JWT with refresh rotation
- Role-Based Access Control (RBAC)
- OTP for sensitive operations
- Session timeout 30 min inactivity
- Device fingerprinting (optional)

9.2 Data Protection

Data Type	Encryption
Password	Argon2id
PII at rest	AES-256
In transit	TLS 1.3
Backups	AES-256
Card data	Tokenization

9.3 Threat Model (STRIDE)

Threat	Mitigation
Spoofing	2FA, JWT signature
Tampering	HTTPS, HMAC
Repudiation	Audit log
Information disclosure	Encryption, RBAC

DoS	Rate limiting, WAF
Elevation of privilege	Least privilege

9.4 Compliance Matrix

Regulation	Requirement	Status
BB ICT Guideline 2023	Security controls	Compliant
PCI-DSS v4.0	Card data	Compliant
ISO 27001	ISMS	Planned
AML/CFT	Transaction monitoring	Compliant
Data retention	7 years	Compliant

10. Testing Requirements

10.1 Test Types

Type	Coverage	Tool
Unit	80%	pytest, Jest
Integration	All APIs	pytest-django
E2E	Critical flows	Cypress/Playwright
Performance	5,000 TPS	Locust/JMeter
Security	OWASP Top 10	OWASP ZAP, Burp
UAT	Business flows	Manual

AI model	Accuracy/FPR	scikit-learn metrics
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10.2 Test Cases (Sample)

TC ID	Description	Expected
TC-01	Register valid user	201 Created
TC-02	Register duplicate email	400 Error
TC-03	Login wrong password 3x	Account locked
TC-04	Transfer insufficient balance	400 + rollback
TC-05	Transfer with valid OTP	Success + SMS
TC-06	Admin approves KYC	Status = APPROVED
TC-07	Fraud score > 0.8	Transaction held
TC-08	JWT expired	401 → refresh
TC-09	SQL injection attempt	Blocked
TC-10	5,000 concurrent transfers	All complete < 5s

10.3 Acceptance Criteria

- All Must-priority FRs pass
- 0 critical/high vulnerabilities
- P95 latency < 2s under load
- UAT sign-off by business

11. Deployment & Operations

11.1 Environments

Env	Purpose	URL
Dev	Development	dev.ibs.local
Staging	Pre-prod testing	staging.ibs.bank.com
Prod	Live	ibs.bank.com

11.2 CI/CD Pipeline

Developer → Git push → GitHub Actions

- |
- | — Lint + Unit test
- | — Build Docker image
- | — Push to Azure Container Registry
- | — Deploy to Staging
- | — Integration + E2E test
- | — Manual approval → Deploy to Prod

11.3 Monitoring & Alerting

- Application Insights: latency, errors, throughput
- Alerts: error rate > 1%, latency > 3s, CPU > 80%
- Slack/Email notifications
- Dashboards for ops team

11.4 Backup & DR

- Daily full backup (Azure Backup)
- Cross-region replication
- DR drill: quarterly

- RTO 1h, RPO 15min

11.5 Support SLA

Severity	Response	Resolution
P1 (Critical)	15 min	4 hours
P2 (High)	1 hour	24 hours
P3 (Medium)	4 hours	5 days
P4 (Low)	1 day	10 days

12. Appendices

Appendix A — Glossary

(উপরে Section 1.3 এ দেওয়া)

Appendix B — Assumptions Log

ID	Assumption	Risk if False
A-01	Core banking API available	Manual reconciliation
A-02	NID API uptime 99%	Manual KYC
A-03	SMS gateway ready	OTP via email

Appendix C — Acronyms

(উপরে দেওয়া)

Appendix D — Traceability Matrix

Req ID	Design	Code	Test
FR-UM-01	Registration flow	users/views.py	TC-01, TC-02
FR-TXN-01	Transfer flow	transactions/views.py	TC-04, TC-05
FR-KYC-02	Approval workflow	kyc/views.py	TC-06
AI-FRD-01	AI service	ai-service/fraud.py	TC-07

Appendix E — Sign-off

Role	Name	Signature	Date
Project Sponsor			
Business Analyst			
Tech Lead			
QA Lead			
Security Officer			
Compliance Officer			



Document Control

Version	Date	Author	Change
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0.1	[date]	[you]	Initial draft
0.5	[date]	[you]	Added AI section
1.0	[date]	[you]	Final for review